HOLY CROSS COLLEGE

Life to the Full

12 PHYSICS

MOMENTS & EQUILIBRIUM

MOMENT (TORQUE)

A moment (M) or torque (τ) is a measure of the turning effect of a force about a point.

It is always applied to a ***rigid structure*** pivoting about a given point.
i.e. It doesn't apply to ropes or wires which aren't rigid.

The magnitude of a moment is given by:

Use either.

$$\boxed{M(\tau) = Fr}$$

$\tau = Fr$

where M = the moment or torque (Nm)
F = the force causing the turning effect (N)
r = the perpendicular distance from the point of rotation to the line of action of the force (m)

EXTENSION

Torque and moment of a force

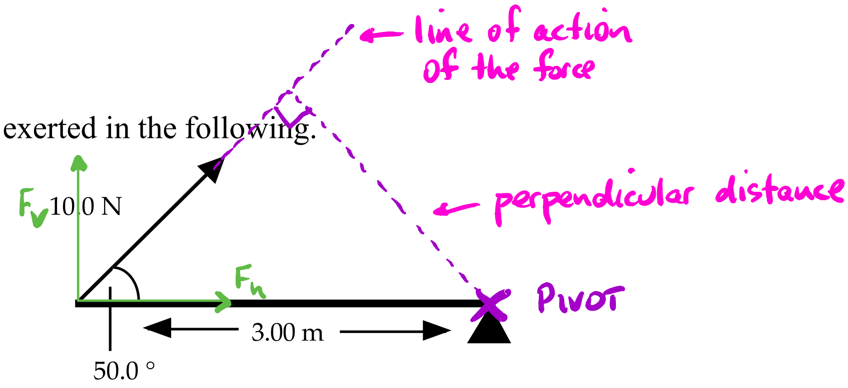
The terms 'torque' (τ) and 'moment of a force' (M), which is usually shortened to moment, are terms that are used interchangeably in physics at the high school level.

The difference between these terms is in the conventions of their use. Typically, torque is used for dynamic problems in which there is an angular acceleration, and so the speed at which an object is rotating is changing. Moments are typically used in static problems, often as reaction forces or internal forces in an object, and are balanced so that there is no angular acceleration. An example of how a torque might be used would be in the force applied by an axle to the wheel of a car to increase its speed of rotation. A moment might be used when calculating the reaction force at the attached point for a beam that is attached to a wall at one end and has a force applied on the other.

Both torque and moment are calculated using the equation $\tau = rF$.

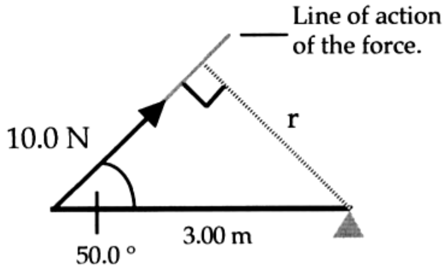
Examples

1. Calculate the moment exerted in the following.



There are ***two*** ways to solve the above problem.

(i) Find r - perpendicular distance to the line of action of the force.



$$\sin 50.0° = \frac{r}{3.00}$$

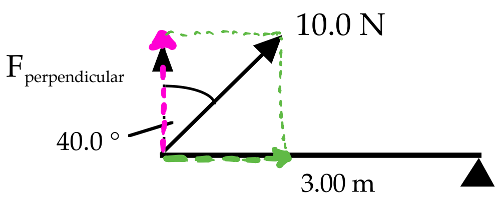
$$\Rightarrow \quad r = 2.298 \text{ m}$$

$$M = Fr$$
$$= (10.0)(2.298)$$
$$= 22.98 \text{ Nm}$$

$\therefore$ <u>M = 23.0 Nm clockwise</u>

Put a direction in.

(ii) Find the perpendicular component of the force (***preferred method***).



$$M = Fr$$
$$= (10.0 \cos 40.0°)(3.00)$$
$$= 22.98 \text{ Nm}$$

$\therefore$ <u>M = 23.0 Nm clockwise</u>

2.

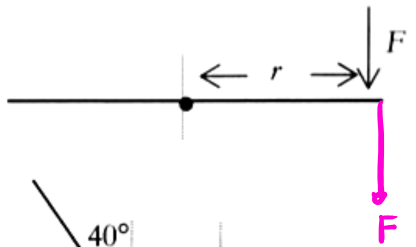
A student riding his bicycle is able to exert a force of 125 N on his bike pedal by pushing directly downwards on it. The pedals are initially in a horizontal position as in Figure 4.3. The centre of the pedal is exactly 17.0cm from the axis of rotation. Determine:

(a) the torque created in this position;

(b) the torque created when the pedal arm makes an angle of 40° with the vertical.

(a) F = 125 N, r = 0.170 M, M = ?

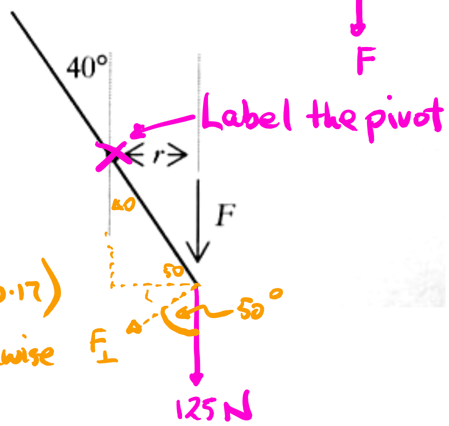
$M = rF$
$= (125)(0.170)$
$= 21.2$ Nm



(b) F = 125 N, r = ?, M = ?, = 40°

$r = (0.17)(\text{Sin } 40°)$
$= 0.109$ m

$\therefore M = (125)(0.109)$
$= 13.7$ Nm



$M = (125 \cos 50.0°)(0.17)$
$= 13.7$ Nm clockwise

EQUILIBRIUM

Definitions

TRANSLATIONAL MOTION: movement in a straight line.

ROTATIONAL MOTION: spins about an axis or fixed point.

A body can possess one or both of these motions.

If a body moves with constant velocity or is at rest, it is not accelerating.

i.e. $\mathbf{\Sigma F = 0}$

∴ It is in ***mechanical equilibrium***.

Doesn't apply to ropes or cables.

If a rigid body does not rotate about a given point, then all of the moments acting on it must balance.

i.e $\mathbf{\Sigma M = 0}$

∴ It is in ***rotational equilibrium***.

For a rigid body to be in equilibrium, it must be in ***mechanical equilibrium and rotational equilibrium***.

Conditions Of Equilibrium

1. The resultant of all forces acting on a body must be zero.

   i.e $\mathbf{\Sigma F = 0}$

   => $\mathbf{\Sigma F_v = 0}$ $\mathbf{\Sigma F_h = 0}$ Use components to solve.

2. If a body has a number of coplanar forces acting on it, then the algebraic sum of their moments about any point in the plane is zero.

   i.e. $\mathbf{\Sigma M = 0}$

   Pick a pivot point where there is an "unknown force".

This second condition for equilibrium is known as the ***Principle of Moments***.

It can be written as:

$$\boxed{\begin{array}{c}\mathbf{\Sigma \text{ clockwise moments} = \Sigma \text{ anticlockwise moments}} \\ \\ \mathbf{\text{i.e.} \quad \Sigma CM = \Sigma ACM}\end{array}}$$

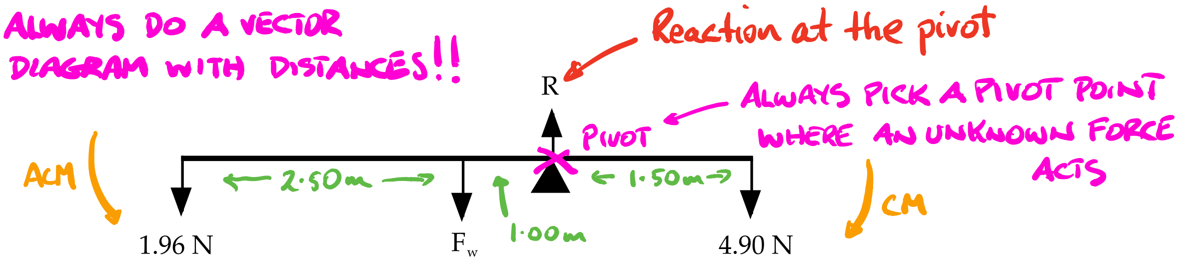
Examples

Same throughout => C. of M is in the middle.

1. **See-Saws**

A uniform beam 5.00 m long has a mass of 0.200 kg hung at one end and a mass of 0.500 kg hung at the other end. When the beam is pivoted at a point 1.50 m from the larger mass, the beam just balances. Calculate:

(a) the weight of the beam.
(b) the reaction force exerted by the fulcrum at the pivot point.



$$
\begin{aligned}
\text{(a)} \qquad \Sigma CM &= \Sigma ACM \\
\Rightarrow \quad (4.90)(1.50) &= (F_w)(1.00) + (1.96)(3.50) \\
\Rightarrow \quad F_w &= 0.490\ \text{N}
\end{aligned}
$$

∴ <u>Weight of the beam = 0.490 N</u>

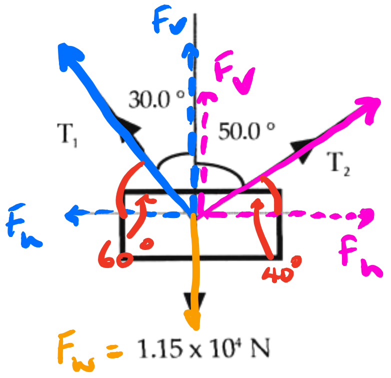
$$
\begin{aligned}
\text{(b)} \qquad \Sigma F &= 0 \qquad \Sigma F_v = 0 \\
\Rightarrow \quad R &= 1.96 + 0.490 + 4.90 \\
&= 7.35\ \text{N}
\end{aligned}
$$

∴ <u>Reaction force = 7.35 N upwards</u>

2. **Ropes And Cables**

(No rigid body is present so moments are *not* considered.)

Two cranes are used to lift a sea-container from the hold of a ship as shown. The container weighs 1.15×10^4 N and moves at constant velocity vertically. Calculate the tension in each cable.



$$\Sigma F_v = 0$$

$$\Rightarrow \quad T_1 \cos 30.0° + T_2 \cos 50.0° = 1.15 \times 10^4 \quad \textbf{[Eqn 1]}$$

$$\Sigma F_h = 0$$

$$\Rightarrow \quad T_1 \cos 60.0° = T_2 \cos 40.0°$$

$$\Rightarrow \quad T_1 = \frac{T_2 \cos 40.0°}{\cos 60.0°} \quad \textbf{[Eqn 2]}$$

Substitute

Substitute Eqn 2 into Eqn 1.

$$\Rightarrow \quad \frac{T_2 \cos 40.0°}{\cos 60.0°} \,.\, \cos 30.0° + T_2 \cos 50.0° = 1.15 \times 10^4$$

$$T_2 \left[\frac{\cos 40.0°}{\cos 60.0°} \cdot \cos 30.0° + \cos 50.0° \right] = 1.15 \times 10^4$$

$$T_2 = \frac{1.15 \times 10^4}{1.9696}$$

$$= 5.839 \times 10^3 \text{ N}$$

Substitute for T_2 in Eqn 2.

$$\Rightarrow \quad T_1 = \frac{5.839 \times 10^3 \cos 40.0°}{\cos 60.0°}$$

$$= 8.946 \times 10^3 \text{ N}$$

$$\therefore \quad \underline{T_1 = 8.95 \times 10^3 \text{ N},\ T_2 = 5.84 \times 10^3 \text{ N}}$$

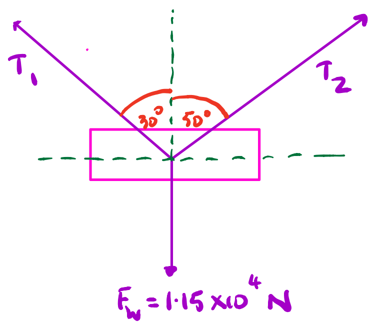
Note

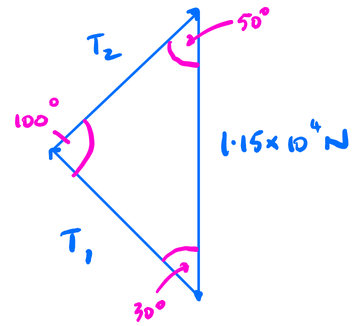
This problem can be solved by having the three vectors acting form **a *"null-vector triangle"***. That is, the vectors add head-to-tail to give a complete triangle.

This is possible since $\Sigma F = 0$.

The following example shows both methods.

Choose the method that you find easiest to understand and use!



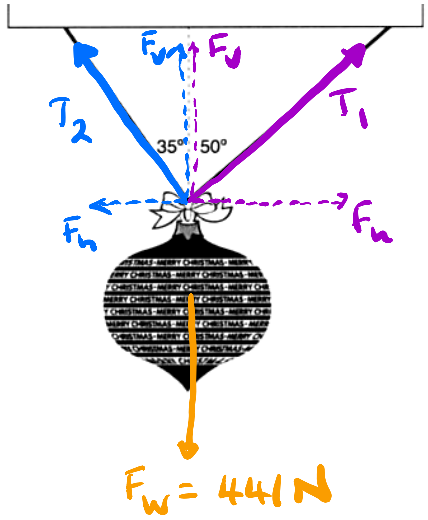


$$\frac{T_1}{\sin 50°} = \frac{1.15 \times 10^4}{\sin 100°} \qquad \frac{T_2}{\sin 30°} = \frac{1.15 \times 10^4}{\sin 100°}$$

$\Rightarrow T_1 =$ ________ N $\qquad \Rightarrow T_2 =$ ________ N

Cables supporting a load

A large Christmas decoration weighing 441 N is being supported by two cables as shown. Determine the tension in each cable.



A vector diagram, approximately to scale is essential. Note that T_2 is likely to be greater than T_1 (Why?).

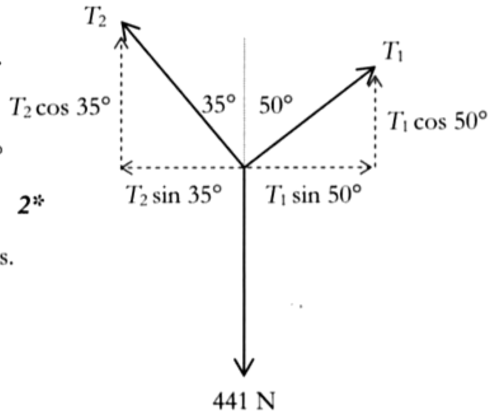
There are two methods of solving this problem.

- **Method A**
  Equate vertical forces and horizontal forces then solve the two simultaneous equations.
- **Method B**
  Use the triangle addition of forces to give a null (zero) vector.

Using method A

$$\sum F_y = 0 \qquad \therefore T_2 \cos 35^\circ + T_1 \cos 50^\circ = 441 \text{ N}$$

$$0.819T_2 + 0.643T_1 = 441 \qquad \textbf{\textit{1*}}$$



Also

$$\sum F_x = 0 \qquad T_2 \sin 35^\circ = T_1 \sin 50^\circ$$

$$0.574T_2 = 0.766T_1 \qquad \textbf{\textit{2*}}$$

Solve ***1**** and ***2**** by eliminating one of the unknowns.

From ***2**** we get $T_2 = \dfrac{0.766}{0.574} = 1.334\ T_1$

substituting into ***1**** we get

$$(0.819)(1.334\ T_1) + 0.643\ T_1 = 441$$

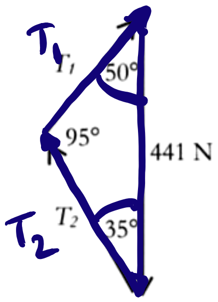
Hence $\qquad T_1 = 254$ N

substitute the value of T_1 into ***1**** and we get

$$0.819\ T_2 + (0.643)(254) = 441$$

$$T_2 = 339 \text{ N}$$

Using method B



Since $\sum F = 0$ simply add the three forces as a triangle of forces giving a null vector. To avoid confusion, always begin by drawing the weight force vector as it is directly down.

Then systematically add the two tension vectors to complete a triangle where the tail of each vector is on the head of the previous one. Be careful not to change the actual orientation of each vector (i.e. the direction it points). In determining the angles of the triangle it's best to work out the angles that the tension forces make with the vertical first.

To solve use the sine rule.

$$\frac{T_1}{\sin 35^\circ} = \frac{T_2}{\sin 50^\circ} = \frac{441}{\sin 95^\circ}$$ ✓

$$\therefore\ T_1 = 441\frac{\sin 35^\circ}{\sin 95^\circ} \quad , \quad T_2 = 441\frac{\sin 50^\circ}{\sin 95^\circ}$$

$$= 254 \text{ N} \qquad\qquad = 339 \text{ N}$$

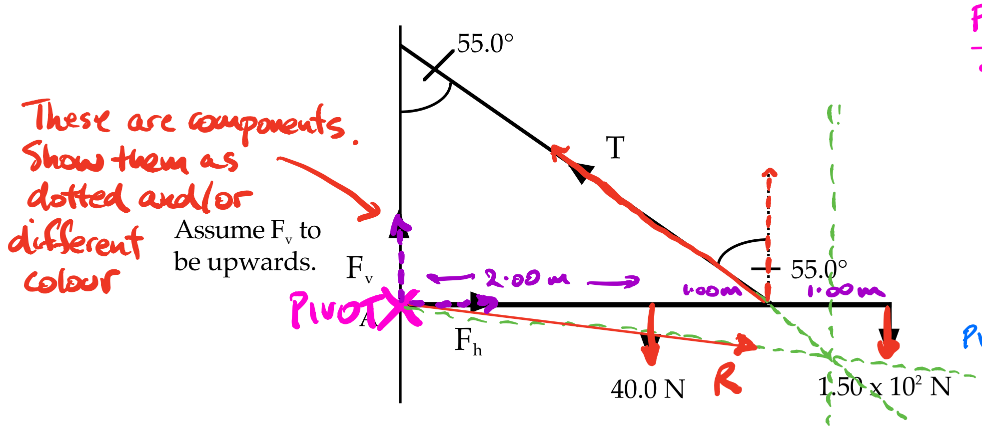
Method B is far superior!

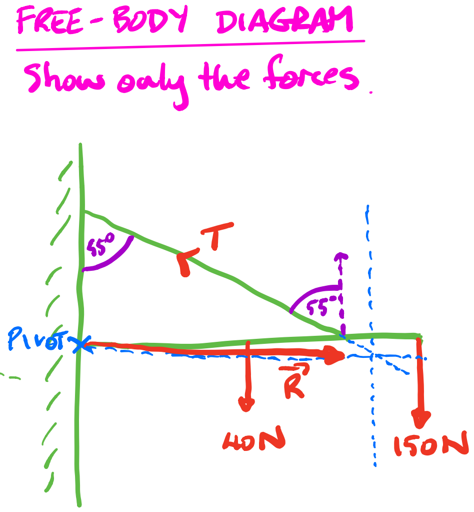
3. **Signs/Drawbridges**

A sign weighing 1.50 x 10^2 N is hung at the end of a 4.00 m long uniform steel bar weighing 40.0 N. The bar is held horizontal by a wire attached 1.00 m from the sign and forming an angle of 55.0° to the wall. Calculate:

(a) the tension in the wire.

(b) the reaction force at the pivot of the wall onto the bar.





STEP2 ΣF=0 (a)

$$\Sigma F_v = 0$$

$$\Rightarrow \quad T\cos 55.0° + F_v = 1.90 \times 10^2 \quad \textbf{[Eqn 1]}$$

$$\Sigma F_h = 0$$

$$\Rightarrow \quad T\cos 35.0° = F_h \quad \textbf{[Eqn 2]}$$

Step 1 ΣCM=ΣACM Take A as pivot.

$$\Sigma CM = \Sigma ACM$$

$$\Rightarrow \quad (40.0)(2.00) + (1.50 \times 10^2)(4.00) = (T\cos 55.0°)(3.00)$$

$$\Rightarrow \quad T = 3.952 \times 10^2 \text{ N}$$

$\therefore$ <u>Tension in the wire = 3.95 x 10^2 N</u>

(b) Substitute for T in Eqn 1 and Eqn 2.

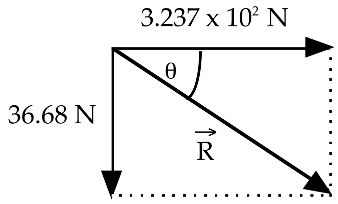
Eqn 1 $\Rightarrow$ $3.952 \times 10^2 \cos 55.0° + F_v = 1.90 \times 10^2$

$\Rightarrow$ $F_v = -3.668 \times 10^1$ N

Take note. (means F_v is down, not up)

Eqn 2 $\Rightarrow$ $3.952 \times 10^2 \cos 35.0° = F_h$

$\Rightarrow$ $F_h = 3.237 \times 10^2$ N



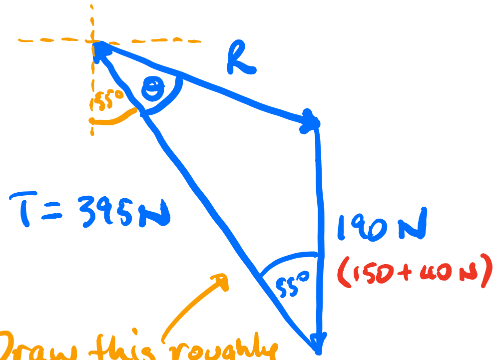
$$\vec{R} = \sqrt{(36.68)^2 + (3.237 \times 10^2)^2}$$
$$= 3.258 \times 10^2 \text{ N}$$

$$\tan\theta = \frac{36.68}{3.237 \times 10^2}$$

$$\Rightarrow \theta = 6.465°$$

$\therefore$ <u>$\vec{R} = 3.26 \times 10^2$ N at 6.46° below the horizontal and away from the wall.</u>

NULL-VECTOR



Draw this roughly to scale.

$$\vec{R} = \sqrt{(395)^2 + (190)^2 - 2(395)(190)\cos 55.0°}$$

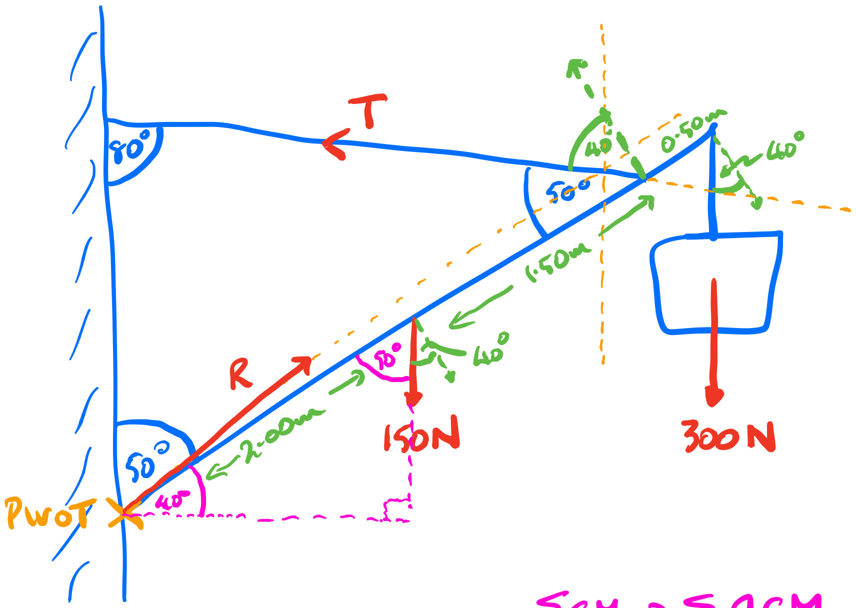
$$= \underline{326} \text{ N}$$

$$\frac{326}{\sin 55.0°} = \frac{190}{\sin\theta}$$

$$\Rightarrow \theta = 28.5°$$

$\therefore$ <u>$\vec{R}$ = 326 N at 83.5° to the vertical downwards</u>

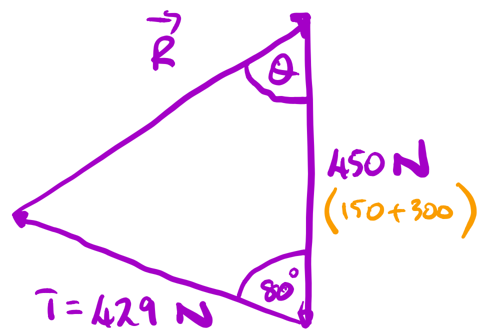
Spare page



$\Sigma CM = \Sigma ACM$

$\Rightarrow (150\cos 40°)(2.00) + (300\cos 40°)(4.00) = (T\cos 40°)(3.50)$

$\Rightarrow T = \underline{429}\ N$



Draw the 2 known vectors roughly to scale.

$\vec{R} = \sqrt{(450)^2 + (429)^2 - 2(450)(429)\cos 80°}$

$= \underline{565}\ N$

$\frac{(429)}{\sin\theta} = \frac{(R)}{\sin 80°}$

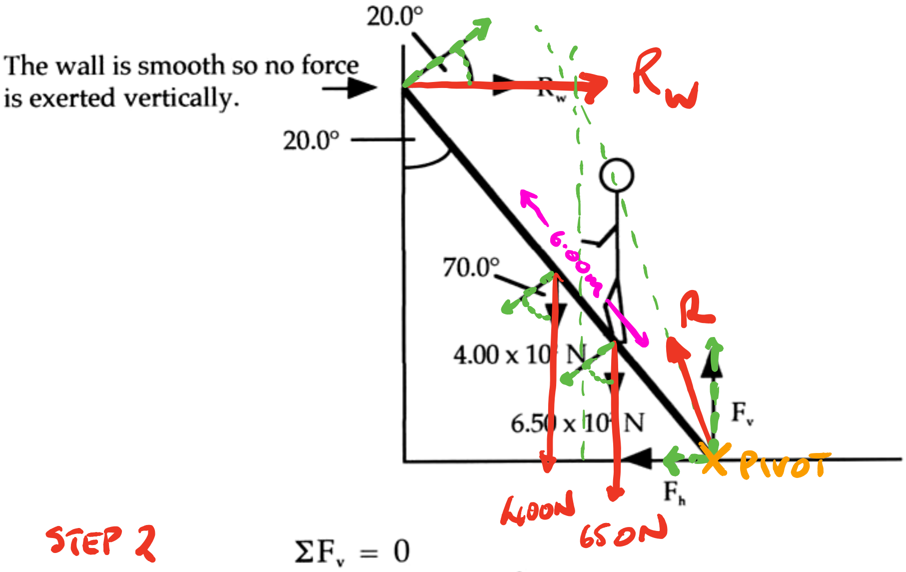
$\Rightarrow \theta = \underline{48.4}°$

$\therefore \vec{R} = \underline{565}$ N at $\underline{48.4}°$ to the vertical upwards

4. **Ladder Against A Wall**

One end of a uniform ladder 6.00 m long and weighing 4.00×10^2 N rests against a smooth wall, forming an angle of 20.0° to it. A person weighing 6.50×10^2 N is standing on a rung 2.00 m from the bottom of the ladder. Calculate:

(a) the force exerted by the wall onto the ladder.
(b) the reaction force exerted by the ground onto the base of the ladder.



(a) STEP 2

$$\Sigma F_v = 0$$
$$\Rightarrow \quad F_v = 1.05 \times 10^3 \text{ N}$$

$$\Sigma F_h = 0$$
$$\Rightarrow \quad F_h = R_w \qquad \textbf{Eqn 1}$$

Take the bottom of the ladder as pivot.

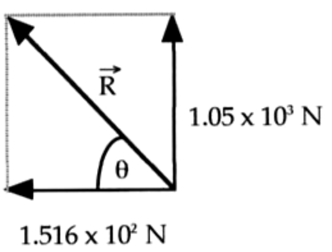
STEP 1

$$\Sigma CM = \Sigma ACM$$
$$\Rightarrow \quad (R_w \cos 20.0°)(6.00) = (6.50 \times 10^2 \cos 70.0°)(2.00) + (4.00 \times 10^2 \cos 70.0°)(3.00)$$
$$\Rightarrow \quad R_w = 1.516 \times 10^2 \text{ N}$$

$\therefore$ $\underline{R_w = 1.52 \times 10^2 \text{ N away from the wall.}}$

(b) Substitute for R_w in Eqn 1.

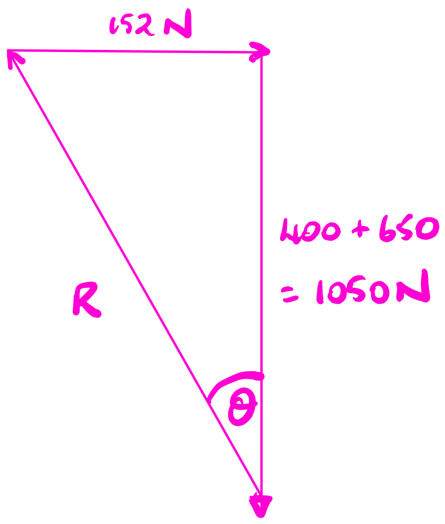
$$\Rightarrow \quad F_h = 1.516 \times 10^2 \text{ N}$$



$$R = \sqrt{(1.05 \times 10^3)^2 + (1.516 \times 10^2)^2}$$
$$= 1.061 \times 10^3 \text{ N}$$

$$\tan\theta = \frac{1.05 \times 10^3}{1.516 \times 10^2}$$
$$\Rightarrow \quad \theta = 81.78°$$

$\therefore$ $\underline{\vec{R} = 1.06 \times 10^3 \text{ N at } 81.8° \text{ above the ground.}}$



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$$R = \sqrt{(1050)^2 + (152)^2}$$

$$= \underline{1.06 \times 10^3}\ N$$

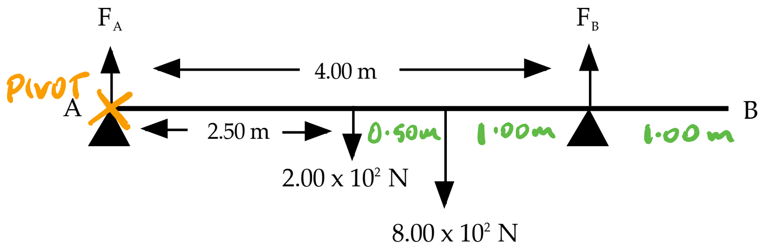
$$\tan\theta = \frac{152}{1050}$$

$$\Rightarrow \theta = \underline{8.2}^\circ$$

$\therefore$ R = ____ N at ____° to the vertical

5. **Bridges**

A rigid uniform plank 5.00 m long and weighing 2.00×10^2 N is supported by two trestles - one at end A and the other 1.00 m in from end B. If a man weighing 8.00×10^2 N stands 2.00 m away from end B, calculate the force exerted by each trestle on the plank.



STEP 2

$$\Sigma F_v = 0$$

$$\Rightarrow \quad F_A + F_B = 1.00 \times 10^3 \text{ N} \qquad \textbf{Eqn 1}$$

There are no horizontal components.

$$\Sigma F_h = 0$$

No forces exerted at an angle => **DISREGARD**

STEP 1

Take A as pivot.

$$\Sigma CM = \Sigma ACM$$

$$\Rightarrow \quad (2.00 \times 10^2)(2.50) + (8.00 \times 10^2)(3.00) = F_B(4.00)$$

$$\Rightarrow \quad F_B = 7.25 \times 10^2 \text{ N}$$

Substitute for F_B in Eqn 1.

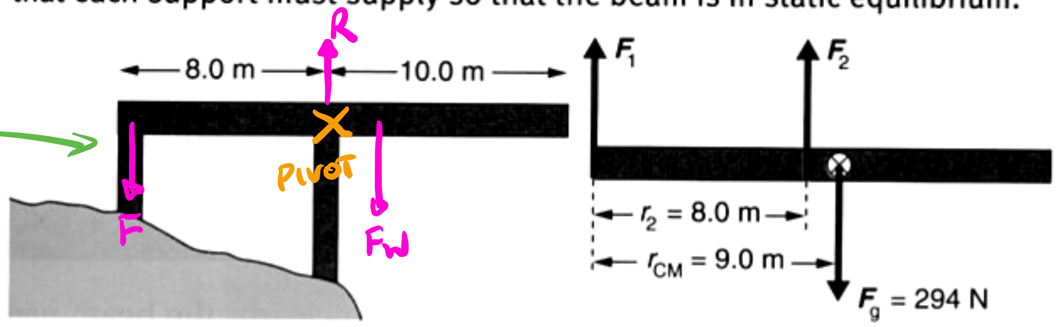
$$\Rightarrow \quad F_A + 7.25 \times 10^2 = 1.00 \times 10^3$$

$$\Rightarrow \quad F_A = 2.75 \times 10^2 \text{ N}$$

$\therefore$ $\underline{F_A = 2.75 \times 10^2 \text{ N upwards, } F_B = 7.25 \times 10^2 \text{ N upwards}}$

A uniform cantilever beam 18.0 m long is used as a viewing platform. It extends 10.0 m beyond two supports which are 8.0 m apart. If the beam has a mass of 30 kg, determine the magnitude and direction of the force that each support must supply so that the beam is in static equilibrium.

Acts down to provide ACM.



Solution

The beam is 18.0 m long, so its centre of gravity is 9.0 m from each end. We will use the up is positive sign convention. A free-body force diagram shows the location and direction of all the forces (b). Since the beam is in rotational equilibrium, the sum of the torques will be zero. If the axis of rotation is located at the left end where $\boldsymbol{F}_1$ is acting, then there is no torque due to $\boldsymbol{F}_1$.

$m_{beam} = 30.0$ kg

$r_{\perp \text{ com beam}} = 9.00$ m

$r_{\perp F_2} = 8.00$ m

$g = 9.80 \text{ N kg}^{-1}$

$$\Sigma\tau_{cw} = \Sigma\tau_{cw}$$

$$F_{beam} r_{\perp \text{ com beam}} = F_2 r_{\perp F_2}$$

$$F_2 = \frac{F_{beam} r_{\perp \text{ com beam}}}{r_{\perp F_2}}$$

$$= \frac{(30.0)(9.80)(9.00)}{8.00}$$

$$= 331 \text{ N upwards}$$

Rather than using the equilibrium condition for torque again, we can find $\boldsymbol{F}_1$ using the condition for translational equilibrium:

$\updownarrow^{+}_{-}$

$F_2 = +331$ N

$F_{beam} = -294$ N

$$\Sigma F = 0$$

$$\therefore F_1 + F_2 + F_{beam} = 0$$

$$F_1 + (331)(-294) = 0$$

$$F_1 + (36.8) = 0$$

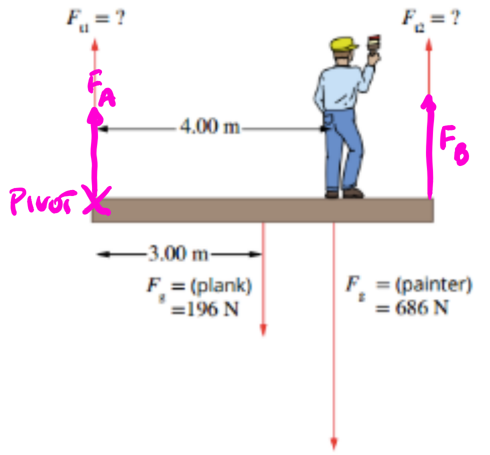
$$F_1 = -36.8 \text{ N}$$

$$= 36.8 \text{ N downwards}$$

CALCULATING STATIC EQUILIBRIUM WITH TWO UNKNOWNS

A 70.0 kg painter stands 4.00 m from the end of a 6.00 m long plank that is supported by a rope at each end. The plank has a mass of 20.0 kg. Determine the tension on the left-hand rope (F_{t1}). Use $g = 9.80\ \mathrm{N\,kg^{-1}}$ in your calculations where required.

BRIDGE PROBLEM



$\Sigma CM = \Sigma ACM$

$\Sigma F_v = 0$

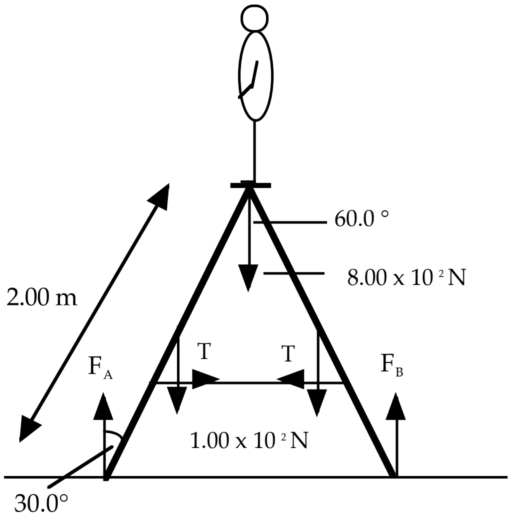
Thinking	Working
Identify the variables involved and state them in their standard form.	$m_{pl} = 20.0\ \mathrm{kg}$ $m_{pa} = 70.0\ \mathrm{kg}$ $r_{\perp F_{t1}-F_{t2}} = 6.00\ \mathrm{m}$ $r_{\perp c-F_{t2}} = 3.00\ \mathrm{m}$ $r_{\perp pa-F_{t2}} = 2.00\ \mathrm{m}$ $g = 9.80\ \mathrm{N\,kg^{-1}}$
Identify the object that is in rotational equilibrium. This is the object on which all the torques are acting.	The object experiencing rotational equilibrium is the plank.
Decide the reference point about which the torques will be calculated. Always choose a point at which one of the unknown forces acts, in order to form an equation with just one unknown. This avoids the need to solve simultaneous equations.	The reference point is the point at which the rope providing the tension force F_{t2} is attached.
Decide which force causes the clockwise torques and which forces cause the anticlockwise torques around the chosen reference point.	The tension force of the left-hand rope on the plank provides the clockwise torque. The force of the painter on the plank provides an anticlockwise torque. The force of gravity on the plank provides another anticlockwise torque.
Apply the equation for rotational equilibrium.	$\Sigma\tau_{clockwise} = \Sigma\tau_{anticlockwise}$
Expand the equation to include each of the torques acting on the plank. The torque of the right-hand rope is not included as it acts through the reference point.	$F_{t1} r_{\perp F_{t1}-F_{t2}} = F_{pl} r_{\perp c-F_{t2}} + F_{pa} r_{\perp pa-F_{t2}}$
Substitute the data into the equation and solve for the unknown force.	$F_{t1} \times 6.00 = 20 \times 9.80 \times 3.00 + 70 \times 9.80 \times 2.00$ $F_{t1} = \dfrac{20 \times 9.80 \times 3.00 + 70 \times 9.80 \times 2.00}{6.00}$ $= \dfrac{588 + 1372}{6.00}$ $F_{t1} = 327\ \mathrm{N}$

6. **Free-Standing Step Ladder**

These types of problems can be very difficult and won't be considered for the WACE examination.

A man weighing 8.00×10^2 N stands on top of a step ladder weighing 2.00×10^2 N as shown. The rope is attached 0.700 m from the bottom of the ladder. Calculate:

(a) the reaction force exerted by the ground (assumed to be smooth).
(b) the tension in the rope holding the sides of the ladder together.



Assume the floor to be smooth.
$\therefore$ F_A and F_B are vertical.

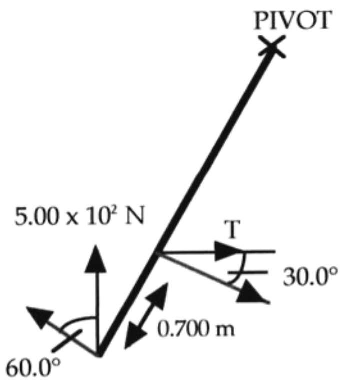
Since the man is on top of the ladder, the force exerted by each side must be equal.
i.e. $F_A = F_B$

(a) $\Sigma F_v = 0$
=> $F_A + F_B = 1.00 \times 10^3$

Since $F_A = F_B$:
$\underline{F_A = F_B = 5.00 \times 10^2 \text{ N}}$

NOTE
Each side would have two feet, meaning that each would exert a force of 2.50×10^2 N upwards.

The tension in the rope "cancels out" horizontally.
i.e. $\Sigma F_h = 0$
=> T = T!



(b) Consider **one side** of the ladder only and take the top as pivot.

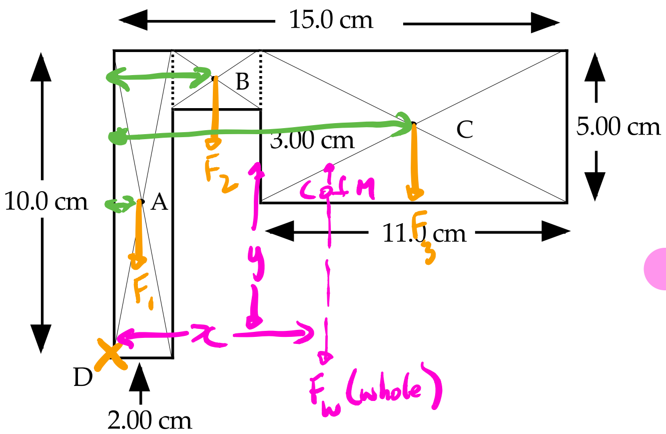
$$\Sigma CM = \Sigma ACM$$
=> $(5.00 \times 10^2 \cos 60.0°)(2.00) = (T \cos 30.0°)(1.30)$
=> $T = 4.441 \times 10^2$ N

$\therefore$ $\underline{T = 4.44 \times 10^2 \text{ N}}$

7. **Lamina**

These types of problems can be very difficult and won't be considered for the WACE examination.

For the following lamina, determine the position of the centre of mass from point D.



The **moment of the whole object** about the given point is equal to the **sum of the moments of the individual parts**.

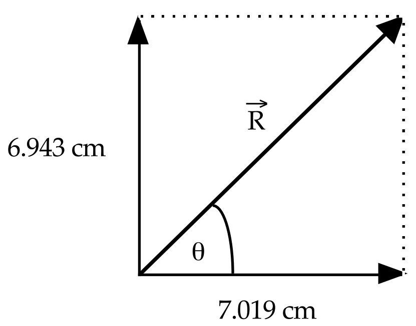
The lamina is assumed to be uniform in thickness.
∴ The **area** of each part gives the **mass** of each part.

The shape is broken up into regular shapes whose centres of mass are easy to find.

Mass (A) = 2.00 x 10.0 = 20.0
Mass (B) = 2.00 x 2.00 = 4.00
Mass (C) = 11.0 x 5.00 = 55.0

X-PLANE M(whole) = M (parts)
=> $(20.0 + 4.00 + 55.0)x = (20.0)(1.00) + (4.00)(3.00) + (55.0)(9.50)$
=> $x = 7.019$ cm

Y-PLANE M(whole) = M(parts)
=> $(20.0 + 4.00 + 55.0)y = (20.0)(5.00) + (4.00)(9.00) + (55.0)(7.50)$
=> $y = 6.943$ cm



$$\vec{R} = \sqrt{(6.943)^2 + (7.019)^2}$$
$$= 9.873 \text{ cm}$$

$$\tan\theta = \frac{6.943}{7.019}$$
$$\Rightarrow \theta = 44.69°$$

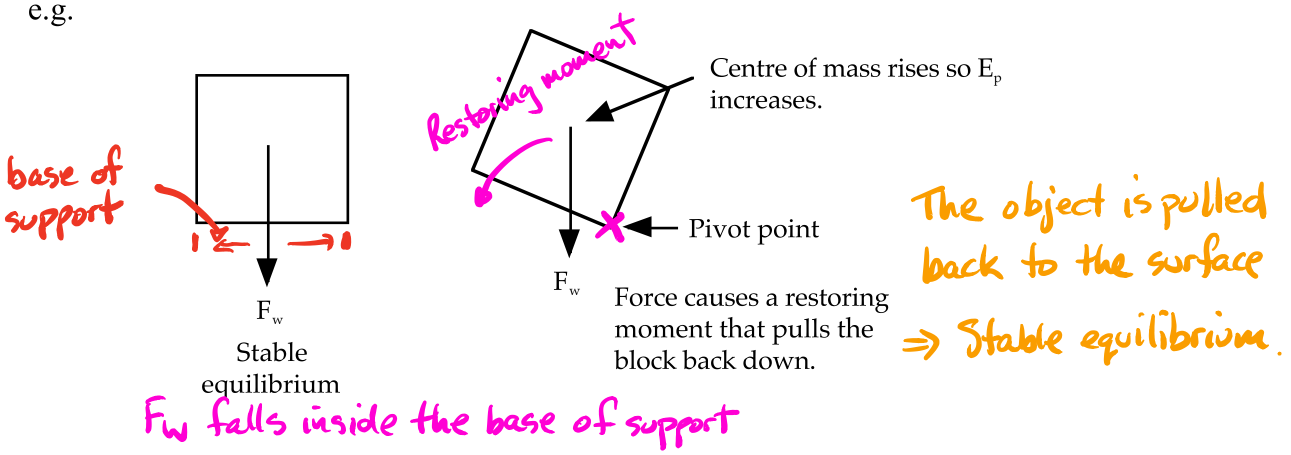
∴ Centre of mass is 9.87 cm from D at 44.7° to the horizontal.

Types Of Equilibrium

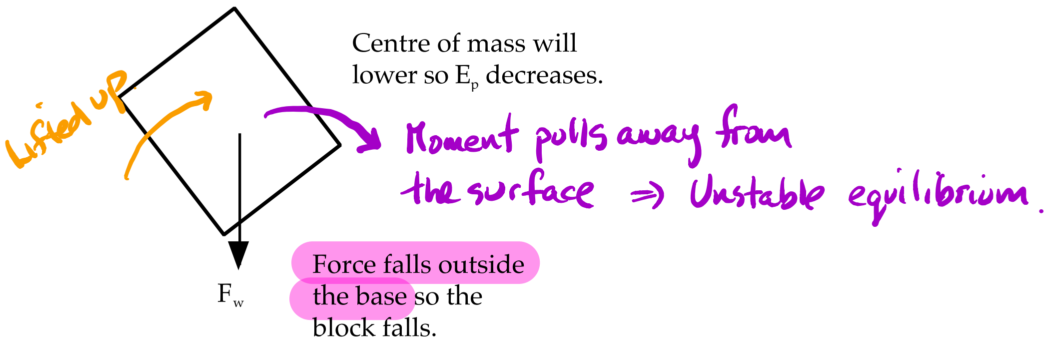
Objects are ***stable*** if their centre of mass falls inside their base of support.

They will ***return to their original position*** after a small displacement.

e.g.

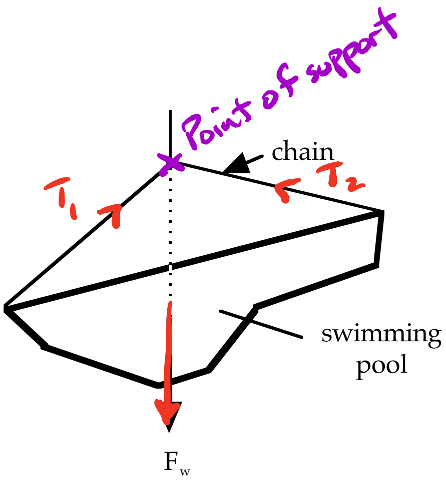


If the block is tipped too far, there is no restoring moment and the block falls over. This is ***unstable*** equilibrium.



Any object that is suspended will have stable equilibrium since its centre of mass will fall directly below the point of support.

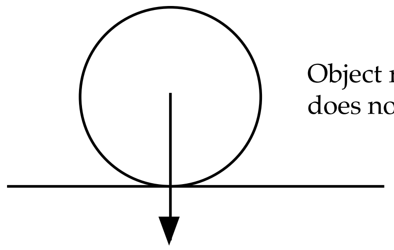
e.g.



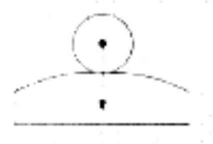
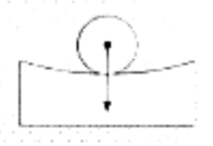
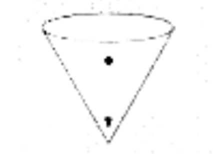
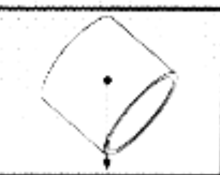
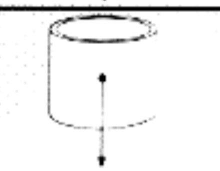
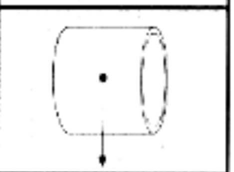
There is a third possible type of equilibrium - ***neutral equilibrium***.

This occurs for objects whose ***centre of mass does not rise nor fall*** when the object is displaced (E_p remains constant).

e.g.



The table below summarises the characteristics of the three types of equilibrium.

	Unstable	Stable	Neutral
Effect of small displacement	continues to move	returns to original position	remains in new position
Position of centre of mass	directly over pivot	creates stabilising torque	always directly over pivot
Potential energy of centre of mass after a small displacement	decreases	increases	remains the same
Example 1 ball on surface			
Examples 2 cone			
Examples 3 cylinder			

There are three types of equilibrium related to the stability of an object:

- **stable equilibrium**—The object will return to its equilibrium position even when a force is applied to it, as long as the centre of mass is not moved outside the **base** of support.
- **unstable equilibrium**—The object will accelerate and will not return to its equilibrium position when a force is applied to it, since the centre of mass is readily moved outside the base of support.
- **neutral equilibrium**—The object will remain stationary no matter where it is placed, as any force applied has no effect on the relationship between the centre of mass and the base or point of support.

Figure 3.2.3 illustrates how this applies to real life objects.

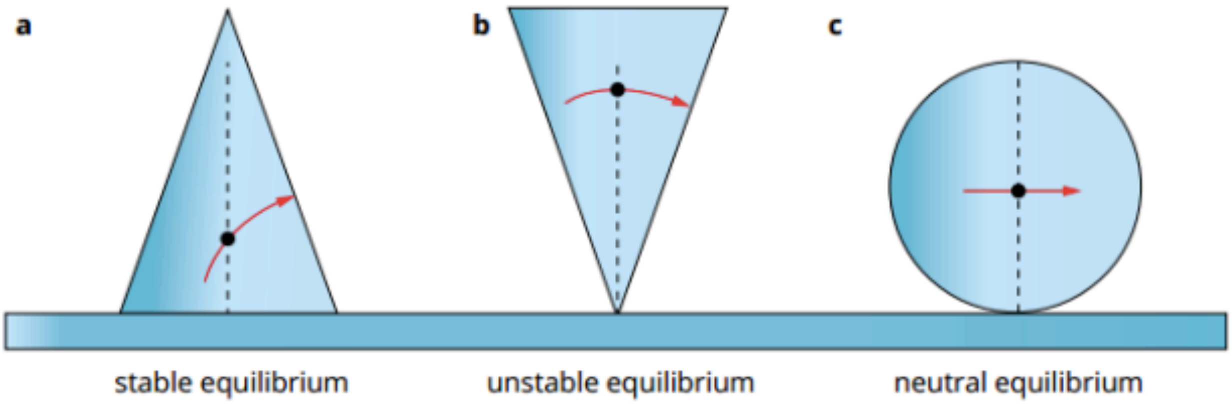


FIGURE 3.2.3 Three objects illustrating the difference between (a) stable, (b) unstable and (c) neutral equilibrium. As (a) has a broad base of support it will return to its original position even when a force is applied to it, as long as the centre of mass isn't moved outside the base of support.

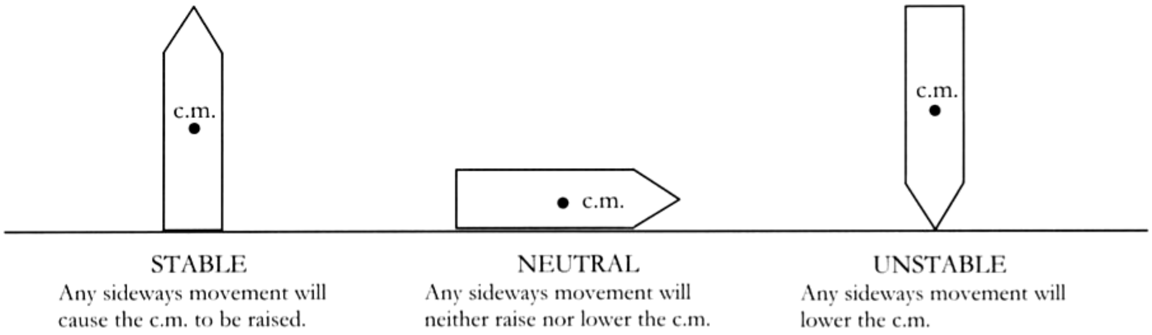
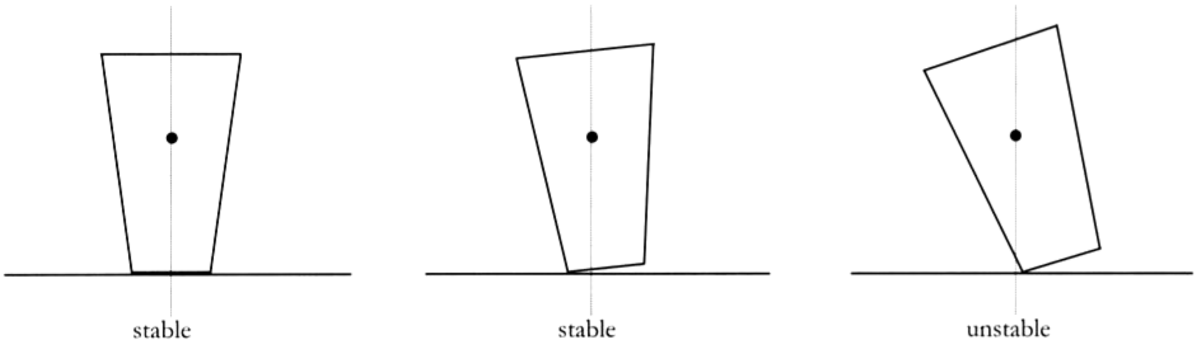


Figure 4.4 The three types of equilibrium

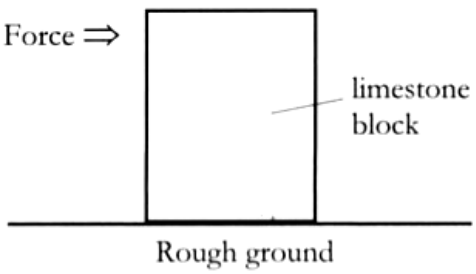
Good stability is generally achieved by keeping the centre of mass as low as possible. Motor cars have a wide base for stability. Tall buildings use large amounts of heavy concrete in the foundations below ground in order to lower the centre of mass. A structure will become unstable and topple if the line of its centre of mass falls outside the objects base.



Example

Will it tip?

A large limestone block 80 cm high and 60 cm wide is to be tipped over by applying a force at the top as shown. The mass of the block is 385 kg and the force being applied is 1500 N. Will it tip over? (Assume no sliding occurs).



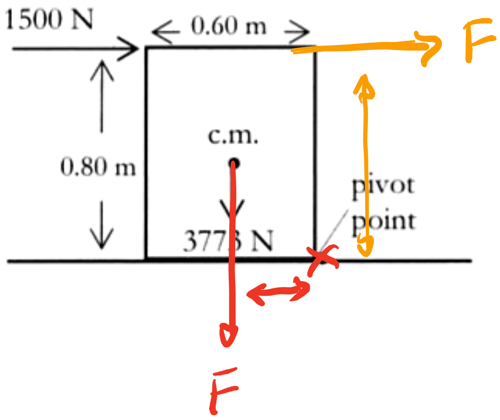
Taking moments about the pivot point - and considering only perpendicular distances

$$\sum \text{c.w.m.} = (1500)(0.80)$$
$$= 1200 \text{ Nm}$$

$$\sum \text{a.c.w.m.} = (3773)(0.30)$$
$$= 1132$$

$$\sum \text{c.w.m.} > \sum \text{a.c.w.m.}$$

$\therefore$ Limestone block will tip.



Physics in action — Centre of mass and stability

Think about an athlete running in a 100 m sprint. In simple terms, the athlete runs in a straight line along the track, and her displacement and velocity at any time can be calculated using the principles discussed in section 1.2. In reality, however, the motion of the various parts of the athlete's body will differ significantly during the run. Her arms and legs move in a complex manner that is not easy to analyse.

The analysis of the motion of complicated systems such as a sprinter or high-jumper can be simplified to the motion of a single point. The mass of the sprinter can be considered to be 'concentrated' into a point which has travelled in a straight line. This single point is called the *centre of mass*. A most important property of the centre of mass is that it will follow a path that is exactly the same as the path of a point particle of the same mass if it were subjected to the same net force.

If a body is uniform in one dimension only (e.g. a straight piece of wire), its centre of mass will lie exactly at the centre. In two dimensions, the centre of mass will be the point which is central for both dimensions. It is even possible for the centre of mass to lie outside the body, as with a doughnut (the centre of mass is in the hole). A person's centre of mass is typically just below the chest, but it will vary with the positions of the arms and legs.

A concept that is closely related to centre of mass is *centre of gravity*. Instead of being a point particle whose motion equates to the whole extended body or system, the centre of gravity is the position from which the entire *weight* of the body or system is considered to act. As a consequence of this, the centre of gravity is the position at which the body will balance. For all practical purposes, the centre of gravity is exactly at the centre of mass. It is only when a body is so large that it is in a non-uniform gravitational field that the centre of gravity no longer coincides with the centre of mass.

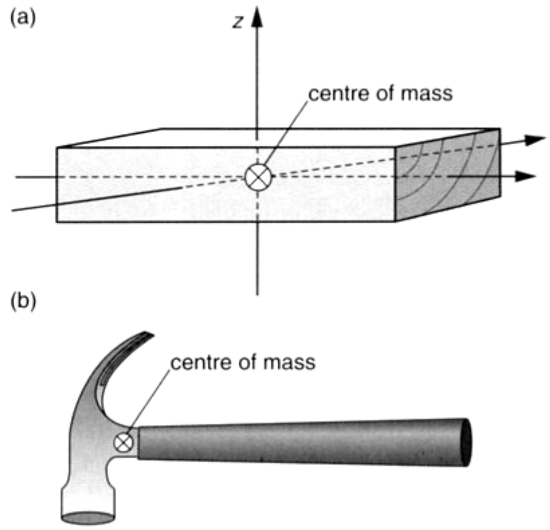


Figure 2.23

The centre of mass for (a) a uniform body in three dimensions and (b) a hammer

In designing structures, engineers and architects want to ensure that balance and stability are maintained. This will depend on the relative positions of the centre of gravity and the base or point of support. When a vertical line downwards from the centre of gravity passes through the base of support, the object is stable. The vertical line from the centre of gravity represents the direction of the force of gravity on the object. In Figure 2.24a the weight of the car passes through the car's support base. The torque acting on the car therefore does not cause the car to tip over. In the case of the truck, however, the weight is directed outside the point (base) of support, so the torque acts to tip the truck over.

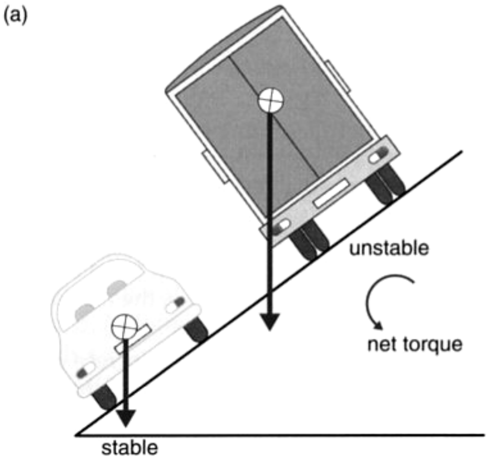


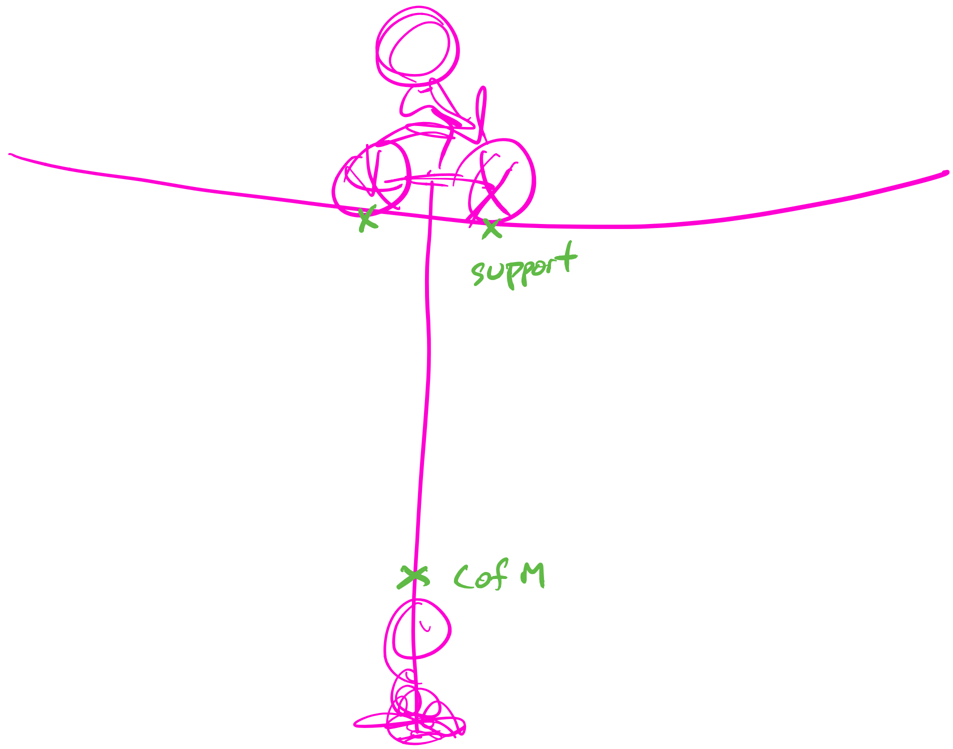
Figure 2.24
(a) The car on the incline is in stable equilibrium, while the heavily laden truck on the same incline could topple. The weight vector is outside the lower point of support for the truck, so there is no reaction force from the road to the higher wheel. (b) Modern four-wheel drives and tractors have inclinometers to warn the driver if the vehicle is in danger of tipping.

The stability of an object or structure can be increased in a number of ways. If the centre of gravity is lowered or the width of the support base is increased, the angle from the centre of gravity to the edge of the base is increased. As a result, the object has to be tipped further to make the force of gravity act outside the support base. Racing cars have a very low centre of gravity to increase their stability when cornering at high speed. In a similar way, training wheels on a child's bicycle widen the support base, making it harder to tip the bicycle sideways.

Figure 2.25
The owner of this cart did not have a good understanding of stability and balanced torques!

The stability of an object or structure can be increased in a number of ways:

- the centre of gravity is lowered
- the width of the support base is increased
- the angle from the centre of gravity to the edge of the base is increased.


support
C of M